Supplementary Material: Sim-to-Real Transferability and Calibration Requirements for Connected-Vehicle Road Roughness Estimation

S1. Analysis plan and deviations

Analysis plan. The estimators and their configurations, the reference simulator V0, the leave-one-route-out split and the metrics were fixed in a written analysis plan before any test-route evaluation, so the comparison of V0-trained with real-trained estimators is confirmatory. The simulator variants, the calibration and equal-budget designs, the fixed-road speed analysis, the spectral comparisons, the band-limited retraining, joint training for all variants and the pass-density analysis were specified in written plans before they were run, but after the test routes had been evaluated; six further analyses were added after some of their results were known (the equal-budget comparison for V1–V8, the per-draw comparison of the calibrated variants with real-only training, the speed scaling of raw vibration in simulation, the paired elasticity differences, the real-only vibration index as a second comparator and the ranking by untrained vibration statistics). All of them are exploratory: every variant, draw and k is reported, and none of them was used to select a configuration. The plans were not registered publicly. Two hypotheses were stated before the corresponding runs. That joint training beats real-only training on average for every variant up to 10 km and stays within 0.02 m/km of it at 20 km holds for V0–V8 and holds with V10. That retraining on the features without vibration above 10 Hz at least halves the spread of the sensor variants' levels while dropping the 10–15 Hz band alone changes it by less than 20 % holds, and V4 stays outside their range (holds). Several analyses had already been computed independently on the same data before the plans were written (joint training for V0, V1, V3 and V5, the band-limited retraining, the paired ranking test, the speed-robustness rows and the no-simulation index), so neither they nor the two hypotheses are confirmatory; where they overlap, our runs reproduce those values (joint training for every fold and draw, and the band-limited retraining, exactly). The game-engine elevation diagnostic of Section S2 was added during the preparation of this manuscript.

Deviations from the plans.

- V7 and V8 use a random half of their passes (seed 7) to match V1 and V2 in size; the full sets are in Table S14.
- A planned aliasing check within one telematics unit was infeasible (TMU#10 logged at 52.6 Hz only on M13 and at 333 Hz only on CPH1 and M3; TMU#12 on M3 had no common segment × speed cell); it was replaced by an all-unit comparison on common cells, which is confounded with device and mounting.
- The keys (segment, task, pass) of the real units are not unique (12 duplicates where a ring-road pass covers a segment twice); predictions are aligned by position and checked against the labels.
- Units shorter than one Welch window have no spectrum and are excluded from the spectral cell means.
- The sensor-robustness experiment was first run with independent draws per perturbation level and then re-run with common random numbers (variance reduction); the first run is kept in `results/sim/robustness_v1_independent_draws.csv`.
- Zero-shot CNNs (three seeds, two simulation sets) and CNNs calibrated with paired drives were limited for compute before their results were known.
- Neural-net metrics are those of the five-seed ensemble prediction; per-seed means are in `results/v2/summary_real_estimators.csv`.
- The telematics configuration of V0 (55–62 Hz, 0.01 g, tilt s.d. 6°) was set from a preliminary inspection of the unlabelled LiRA-CD files. A complete re-measurement gives nominal rates of 52.6 Hz (66 % of passes) or 333 Hz, 0.01 g quantisation in 81 % of passes and tilts with median 10° (maximum 41°). V0 was kept; the matched configurations are the variants V5 and V6.
- The no-simulation comparison also reports the vibration index fitted to V0 simulated units with a level offset (Table S15); the band associations were also computed against the motorway-minus-urban level difference. Neither is used for any claim.

S2. Labels and the game-engine benchmark

IRI labels. On the 7 LiRA-CD route-directions whose profile and IRI files are consistent, our IRI implementation reproduces the IRI reported by the P79 with $r = 1.000$, bias $+0.018$ m/km and MAE 0.019 m/km over 1,520 segments of 100 m (Fig. S1a; an algorithm check, since both are computed from the same laser profiles). For the eighth file (CPH6, direction HH) the

reported IRI cannot be aligned with the file's own profile (best $r = 0.13$ over shifts of ± 3 km, with or without reversal, against $r = 1.000$ at zero shift for two control files); all labels are therefore recomputed from the profiles. A second profilometer (ARAN9000, surveyed on other dates) is compared in Fig. S1c and Table S12.

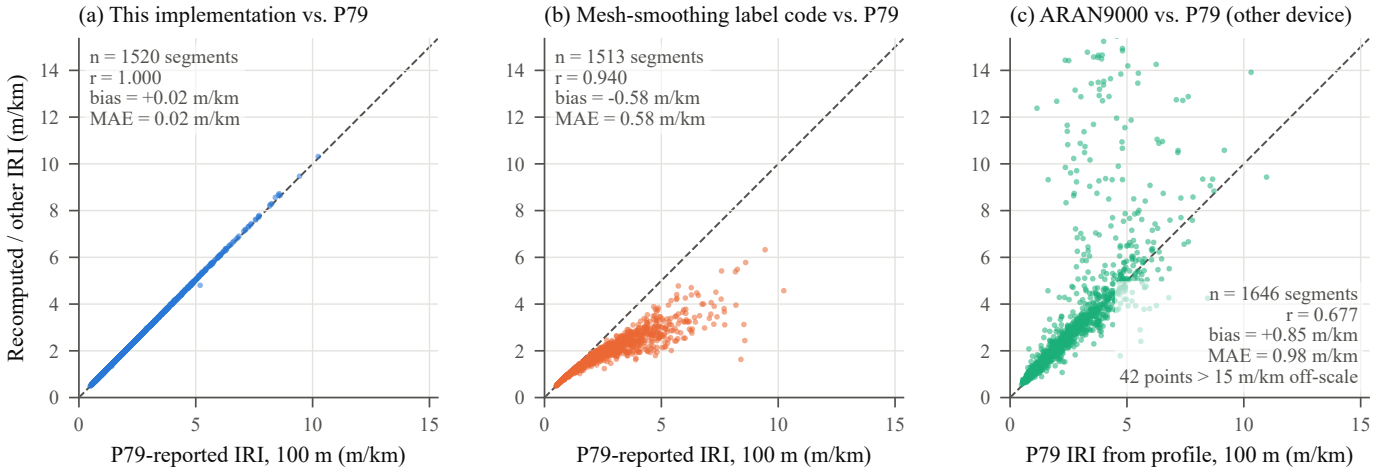


Fig. S1. Label validity on real profiles (LiRA-CD, 100 m segments, mean of both wheel paths). (a) IRI recomputed from the P79 laser profiles with our implementation vs the IRI reported by the P79. (b) The game-engine label code, including its curvature-adaptive mesh smoothing, applied to the same profiles. (c) A second profilometer (ARAN9000, other survey date) vs the P79; off-scale points are counted in the panel and included in the statistics.

Game-engine corpus. The corpus contains 19 manually driven trips (138 km) of 3 BeamNG.tech vehicles on 3 maps. The inertial signals, the position and the ray-cast elevation of the road mesh under each wheel path were polled by separate threads: 48–68 % of consecutive inertial rows are exact sample-and-hold duplicates, and 54–69 % of the elevation readings repeat the previous reading while the car moves (59 % on average, mostly in runs of at most ten readings). Fig. S2 compares the displacement spectra of the logged elevation with those of the measured P79 profiles: the game-engine elevation carries 12 times the power of the urban P79 profiles at wavelengths of 10–30 m and 42–54 times at 0.5–5 m (95 and 269–568 times that of the motorway profiles). The IRI of its non-overlapping 100 m segments averages 12.5 m/km from the logged elevation and 15.6 m/km when the repeated readings are removed before interpolation, so the excess lies in the elevation values, not in their sampling. A curvature-adaptive smoothing designed to remove mesh tessellation lowers the mean to 4.5 m/km, removing 64 % of the summed IRI and changing the condition class of 34 % of the segments. Applied to the measured P79 profiles, which contain no mesh facets, the same smoothing lowers the IRI of urban segments by 28 % on average (bias -1.10 m/km) and that of motorway segments by 6 % (Fig. S1b), so neither the smoothed nor the unsmoothed labels are a valid IRI ground truth.

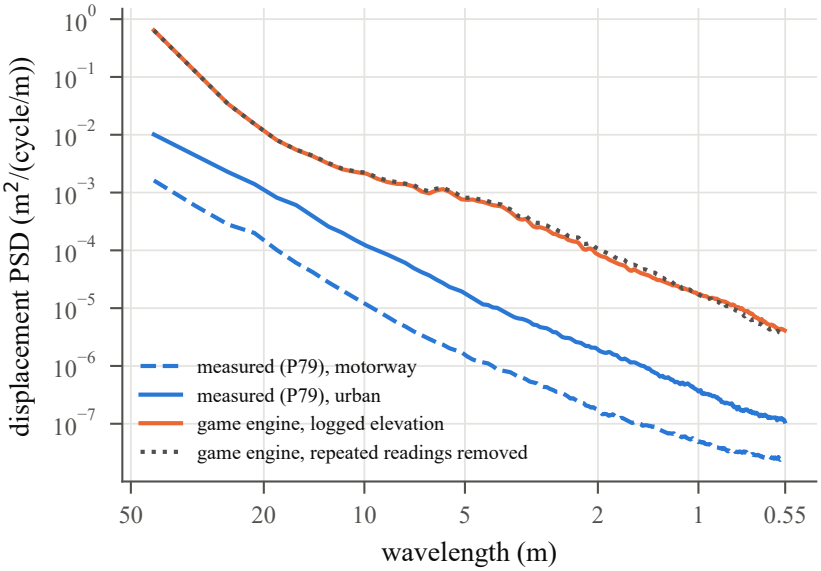


Fig. S2. Displacement power spectral density (Welch, detrended, 0.25 m grid) of the measured P79 profiles of LiRA-CD (laser 5 and 21, gap-free runs) and of the ray-cast mesh elevation of the game-engine corpus (both wheel paths of all trips), as logged and with repeated readings removed before interpolation.

Transfer and generalisation. Trained on the corpus and evaluated on LiRA-CD as in the main paper, LightGBM reaches a within-route MAE of 1.26 m/km with the smoothed and 2.08 m/km with the unsmoothed labels (within-route ρ 0.35 and 0.32; Table S14), and its estimates rise steeply with speed on a fixed road (elasticity $+0.79$ and $+1.39$). The vibration index trained on the same corpus ranks segments within routes far better (ρ 0.83 with the smoothed labels) but misses the level (MAE 0.85 m/km, bias -0.73 m/km; Table S15), the pattern of ranking without level seen with the parametric simulator. Within the corpus, LightGBM trained on two maps reaches an MAE of 2.06 m/km on the third, against 3.01 m/km for the training median, and with both vehicle and map unseen no estimator is substantially better than the training median (MAE 7.60; vibration index 6.88; LightGBM 7.25 m/km, unsmoothed labels).

S3. Simulator

The half-car is discretised exactly under a first-order hold and solved in modal coordinates; the solution deviates by at most $3.9 \cdot 10^{-13}$ (relative) from a direct state-space recursion and by $4.4 \cdot 10^{-9}$ from a fine-step Runge–Kutta integration. The sampled vehicles have body modes at 0.95–2.52 Hz and wheel-hop modes at 9.1–16.1 Hz. Speed is drawn independently of roughness (base speed 20–110 km/h with Ornstein–Uhlenbeck fluctuations and random braking events). Table S1 lists the parameter ranges; the variants and their seeds are defined in `experiments/sim_generate.py`. Besides the variants of the main paper, V10 adds the synthetic profiles of V9 to the measured ones.

TABLE S1
Parameter ranges of the parametric simulator (sensitivity ranges; sources in config/sim_ranges.json).

group	parameter	range	unit	source
vehicle	m_s	1000 - 1800	kg	G, Z
vehicle	r_y	1 - 1.3	m	G (pitch radius of gyration ~ sqrt(a*b))
vehicle	L	2.4 - 3	m	G, Z
vehicle	a_frac	0.4 - 0.5	-	front-heavy passenger cars
vehicle	f_s	1 - 1.6	Hz	G, S
vehicle	rear_freq_ratio	1 - 1.2	-	G (flat-ride tuning)
vehicle	zeta	0.2 - 0.45	-	G, S
vehicle	m_u_per_wheel	30 - 50	kg	G
vehicle	k_t_per_tire	150000 - 300000	N/m	G (wheel hop 10-15 Hz)
vehicle	x_s	-0.5 - 1.2	m	sensor between CG and front axle (LiRA readme: near front axle)
sensor: autopi	rate / LSB / noise / tilt SD	55-62 Hz / 0.01 g / 150-400 ug/rtHz / 6.0 deg	-	M, D
sensor: phone	rate / LSB / noise / tilt SD	90-210 Hz / 0.0006 g / 100-300 ug/rtHz / 12.0 deg	-	D (smartphone IMUs); cradle resonance range is an assumption, treated as sensitivity
speed	base speed	20-110	km/h	design choice (independent of roughness)

S4. Real cars: additional results

Table S2 lists every estimator with its paired difference to real-trained LightGBM (5 km-block intervals), and Table S3 gives every estimator per held-out route. Speed alone ranks segments only weakly ($\rho = 0.39$). LightGBM and the MLP differ negligibly ($+0.004$ [-0.013, $+0.023$] m/km); the deficit of the 1D-CNN ($+0.045$ [$+0.019$, $+0.075$]) lies almost entirely on CPH1 ($+0.14$ m/km there, between -0.02 and $+0.02$ on the other routes), and a log-linear output calibration of the quarter-car inversion on the training routes made it worse. The same telematics units drive several routes, but LightGBM trained only on units that never drove the held-out route is as accurate as on size-matched random subsets (CPH1 0.63 vs 0.63, CPH6 0.69 vs 0.70, M13 0.15 vs 0.15 m/km; Table S8; the M3 fold is not interpretable because only urban training units remain), so vehicle overlap did not measurably favour the real-trained estimators. On the within-road split of Bajic et al. (first 80 % of M3 for training, last 20 % for testing), LightGBM reaches MAE 0.115 m/km and $R^2 = 0.81$ (Table S9); this anchors the benchmark to earlier work on LiRA data but answers a different question. Table S6 gives paired differences at four levels of dependence (segment, 1 km and 5 km blocks, routes; Fig. S3a shows why segment-level intervals are too narrow), Table S7 decomposes pass-level errors into segment, telematics-unit and residual components, Table S10 gives the effect of averaging passes, Table S11 the quarter-car inversion per fold (fitted transmissibility in Fig. S3b) and Table S12 the comparison of the two profilometers. Table S4 compares the within-route ranking of every trained estimator with that of the untrained statistic $\mathrm{RMS}/v^{0.5}$ route by route, and Table S5 gives relative errors by road class.

TABLE S2

REAL CARS (LIRA-CD), LEAVE-ONE-ROUTE-OUT, 100 M SEGMENTS. WITHIN-ROUTE METRICS: MEAN OF THE FOUR HELD-OUT ROUTES (MAIN PAPER, SECTION II); PARTIAL ρ: SPEARMAN CORRELATION WITH IRI GIVEN SEGMENT MEAN SPEED. ΔMAE: PAIRED DIFFERENCE TO THE REAL-TRAINED LIGHTGBM WITH A 95 % CLUSTER-BOOTSTRAP INTERVAL OVER 5 KM BLOCKS (40 CLUSTERS); "ROUTES": NUMBER OF HELD-OUT ROUTES ON WHICH THE ESTIMATOR IS BETTER. NEURAL NETS: 5-SEED ENSEMBLES.

Estimator	Within route			Pooled		ΔMAE vs M3	Routes
	MAE	ρ	partial ρ	MAE	bias	[5 km-block CI]	better
No vibration input							
M0 constant (training median)	1.56	–	–	1.67	−0.64	+1.249 [+1.118, +1.387]	0/4
M0b road-class prior (map only)	0.75	–	–	0.81	−0.20	+0.387 [+0.307, +0.475]	0/4
M1 speed only	0.69	0.39	0.19	0.74	−0.33	+0.324 [+0.252, +0.404]	0/4
Physics-based, calibrated on real training routes							
M2c vibration index	0.62	0.76	0.77	0.67	−0.13	+0.251 [+0.202, +0.306]	0/4
M7 quarter-car inversion	0.46	0.82	0.81	0.49	−0.08	+0.074 [+0.039, +0.105]	0/4
M7 + output calibration	0.54	0.82	0.81	0.58	+0.05	+0.155 [+0.109, +0.201]	0/4
Learned on real cars (3 other routes)							
M3 LightGBM (reference)	0.39	0.83	0.81	0.42	−0.11	ref.	–
M3 LightGBM, no speed input	0.62	0.76	0.72	0.65	−0.24	+0.235 [+0.177, +0.294]	0/4
M4 MLP	0.40	0.83	0.81	0.42	−0.09	+0.004 [−0.013, +0.023]	0/4
M5 1D-CNN	0.43	0.81	0.79	0.47	−0.21	+0.045 [+0.019, +0.075]	1/4
Learned on simulation only (zero-shot)							
V0 LightGBM, measured profiles	0.42	0.83	0.81	0.44	+0.01	+0.019 [−0.003, +0.040]	1/4
V10 LightGBM, measured + synthetic	0.44	0.84	0.82	0.46	−0.04	+0.039 [+0.007, +0.068]	2/4

TABLE S3

PER-ROUTE RESULTS ON REAL CARS (SEGMENT MAE IN M/KM AND WITHIN-ROUTE SPEARMAN ρ; EACH HELD-OUT ROUTE EVALUATED BY THE MODELS TRAINED ON THE OTHER THREE; ρ IS UNDEFINED FOR PREDICTORS THAT ARE CONSTANT WITHIN A ROUTE). SEGMENTS: CPH1 509, CPH6 426, M3 515, M13 228. LAST ROWS: UNTRAINED VIBRATION STATISTICS, WHOSE WITHIN-ROUTE ρ NEEDS NO TRAINING DATA.

	MAE				ρ			
Estimator	CPH1	CPH6	M3	M13	CPH1	CPH6	M3	M13
M0 constant	2.12	2.03	1.31	0.79	–	–	–	–
M1 speed only	1.10	1.11	0.31	0.25	0.48	0.42	0.20	0.45
M2c vibration index	1.03	0.86	0.35	0.25	0.74	0.74	0.86	0.70
M3 LightGBM	0.60	0.70	0.13	0.15	0.88	0.84	0.90	0.72
M3 LightGBM, no speed	1.03	0.74	0.35	0.35	0.80	0.80	0.86	0.57
M4 MLP	0.60	0.70	0.13	0.16	0.88	0.83	0.89	0.72
M5 1D-CNN	0.73	0.68	0.15	0.17	0.87	0.79	0.88	0.71
Sim V0 LightGBM (zero-shot)	0.55	0.73	0.21	0.18	0.88	0.81	0.93	0.73
Sim V10 LightGBM (zero-shot)	0.57	0.69	0.26	0.21	0.88	0.81	0.93	0.73
M0b map-only road-class prior	1.20	1.18	0.34	0.30	–	–	–	–
M7 quarter-car inversion	0.68	0.71	0.27	0.17	0.86	0.82	0.89	0.72
M7 + output calibration	0.70	0.84	0.39	0.22	0.86	0.82	0.89	0.72
untrained: raw RMS	–	–	–	–	0.71	0.65	0.90	0.75
untrained: RMS/$v^{0.5}$	–	–	–	–	0.84	0.79	0.92	0.74
untrained: RMS/v	–	–	–	–	0.86	0.78	0.92	0.73

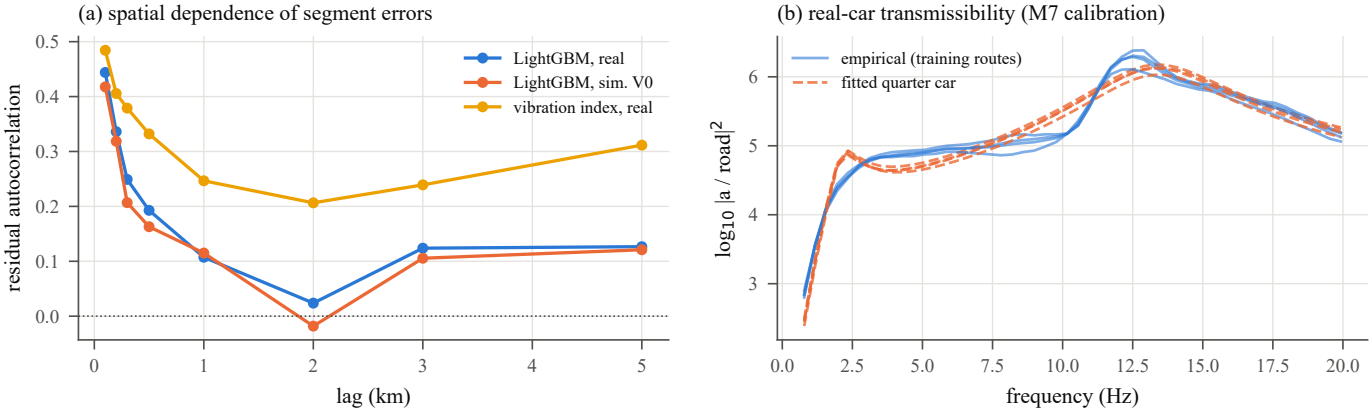


Fig. S3. (a) Autocorrelation of segment-level log errors along the road, by lag. (b) Empirical transmissibility of the real car (acceleration spectrum divided by the road displacement spectrum at the pass speed; training routes of each fold) and the fitted quarter car of the inversion.

TABLE S4

PAIRED TEST OF WITHIN-ROUTE RANKING: SPEARMAN ρ (TOP) AND PARTIAL ρ GIVEN SEGMENT SPEED (MIDDLE) OF EACH ESTIMATOR MINUS THAT OF THE UNTRAINED STATISTIC $\mathrm{RMS}/v^{0.5}$ ON THE SAME SEGMENTS, PER HELD-OUT ROUTE AND MEAN OVER ROUTES, WITH 95 % INTERVALS FROM 5 KM BLOCKS RESAMPLED WITHIN EACH ROUTE (B = 2000). M13 HAS ONLY 6 BLOCKS, SO ITS INTERVALS ARE COARSE. BOTTOM: THE SAME DIFFERENCE IN SPEARMAN ρ AS A POINT ESTIMATE (NO INTERVAL) FOR THE OTHER ESTIMATORS OF TABLE S3 AND FOR LIGHTGBM TRAINED ON THE OTHER SIMULATOR VARIANTS (NEURAL NETS AS 5-SEED ENSEMBLES).

	Estimator	CPH1	CPH6	M3	M13	mean
$\Delta\rho$	real-trained LightGBM	+0.042 [+0.027, +0.062]	+0.044 [+0.032, +0.066]	−0.020 [−0.034, −0.008]	−0.023 [−0.033, −0.010]	+0.011 [+0.004, +0.020]
	V0 LightGBM (zero-shot)	+0.038 [+0.016, +0.064]	+0.017 [+0.002, +0.040]	+0.007 [−0.005, +0.020]	−0.015 [−0.055, +0.023]	+0.012 [−0.001, +0.025]
	V1 LightGBM (zero-shot)	+0.021 [−0.003, +0.047]	+0.002 [−0.023, +0.027]	+0.004 [−0.018, +0.017]	−0.007 [−0.042, +0.036]	+0.005 [−0.008, +0.020]
Δ partial ρ	real-trained LightGBM	+0.026 [+0.019, +0.037]	+0.020 [+0.006, +0.037]	−0.020 [−0.032, −0.010]	−0.023 [−0.040, +0.007]	+0.001 [−0.006, +0.009]
	V0 LightGBM (zero-shot)	+0.019 [−0.004, +0.038]	−0.008 [−0.037, +0.020]	+0.006 [−0.003, +0.015]	−0.016 [−0.063, +0.028]	+0.000 [−0.014, +0.015]
	V1 LightGBM (zero-shot)	−0.001 [−0.022, +0.017]	−0.026 [−0.068, +0.007]	+0.000 [−0.013, +0.011]	−0.009 [−0.038, +0.044]	−0.009 [−0.023, +0.008]
$\Delta\rho$, point estimate	MLP	+0.042	+0.043	−0.032	−0.024	+0.007
	1D-CNN	+0.027	−0.006	−0.034	−0.035	−0.012
	M7 quarter-car inversion	+0.022	+0.030	−0.026	−0.019	+0.002
	V2 LightGBM (zero-shot)	+0.038	+0.022	−0.005	−0.019	+0.009
	V3 LightGBM (zero-shot)	+0.026	+0.013	+0.008	−0.006	+0.010
	V4 LightGBM (zero-shot)	+0.036	+0.025	−0.005	−0.032	+0.006
	V5 LightGBM (zero-shot)	+0.026	−0.016	+0.005	−0.022	−0.001
	V6 LightGBM (zero-shot)	+0.031	+0.014	+0.003	−0.021	+0.007
	V7 LightGBM (zero-shot)	+0.037	+0.023	+0.013	−0.025	+0.012
	V8 LightGBM (zero-shot)	+0.026	+0.021	+0.009	+0.000	+0.014
	M2c vibration index	−0.099	−0.054	−0.061	−0.038	−0.063
	LightGBM without speed inputs	−0.041	+0.007	−0.055	−0.176	−0.066
	M1 speed only	−0.361	−0.373	−0.723	−0.291	−0.437

TABLE S5

RELATIVE ERRORS ON REAL CARS: SEGMENT MAE OVER THE TWO HELD-OUT ROUTES OF EACH ROAD CLASS DIVIDED BY THEIR MEAN IRI (%; ESTIMATORS OF TABLE S2, LEAVE-ONE-ROUTE-OUT).

Estimator	urban	motorway
M0 constant	62	100
M1 speed only	33	25
M2c vibration index	28	28
M3 LightGBM	19	12
M3 LightGBM, no speed	27	30
M4 MLP	19	12
M5 1D-CNN	21	14
Sim V0 LightGBM (zero-shot)	19	17
Sim V10 LightGBM (zero-shot)	19	22
M0b map-only road-class prior	35	28
M7 quarter-car inversion	21	21
M7 + output calibration	23	29

TABLE S6

PAIRED MAE DIFFERENCES (M/KM) AT FOUR LEVELS OF DEPENDENCE: CLUSTER BOOTSTRAP OVER SEGMENTS, 1 KM AND 5 KM BLOCKS (B = 2000), AND THE ROUTE-LEVEL DESCRIPTIVE INTERVAL (MEAN OF THE 4 PER-ROUTE DIFFERENCES, t WITH 3 DF). REFERENCE: REAL-TRAINED LIGHTGBM UNLESS STATED.

Estimator	Diff.	Segment	1 km block	5 km block	Route (t_3)
V0 LightGBM (zero-shot)	+0.019	[+0.008, +0.031]	[+0.002, +0.036]	[−0.003, +0.040]	[−0.06, +0.10]
V10 LightGBM (zero-shot)	+0.039	[+0.026, +0.052]	[+0.019, +0.059]	[+0.007, +0.068]	[−0.07, +0.15]
MLP (per-seed mean)	+0.005	[−0.003, +0.012]	[−0.007, +0.017]	[−0.012, +0.024]	[+0.00, +0.01]
MLP (ensemble)	+0.004	[−0.004, +0.011]	[−0.008, +0.016]	[−0.013, +0.023]	[+0.00, +0.01]
1D-CNN (per-seed mean)	+0.055	[+0.042, +0.068]	[+0.035, +0.076]	[+0.029, +0.086]	[−0.07, +0.16]
1D-CNN (ensemble)	+0.045	[+0.032, +0.059]	[+0.025, +0.066]	[+0.019, +0.075]	[−0.07, +0.15]
speed only	+0.324	[+0.295, +0.351]	[+0.277, +0.373]	[+0.252, +0.404]	[−0.01, +0.60]
vibration index M2c	+0.251	[+0.226, +0.275]	[+0.212, +0.292]	[+0.202, +0.306]	[−0.00, +0.46]
road-class prior (map)	+0.387	[+0.356, +0.418]	[+0.339, +0.438]	[+0.307, +0.475]	[+0.02, +0.70]
road-class prior (map) vs speed only	+0.064	[+0.045, +0.082]	[+0.028, +0.100]	[+0.029, +0.101]	[+0.02, +0.11]

TABLE S7

VARIANCE COMPONENTS OF PASS-LEVEL LOG ERRORS (TWO-WAY DECOMPOSITION: SEGMENT EFFECTS + TELEMATICS-UNIT EFFECTS; SD IN % OF LOG ERROR; UNIT SD WITH A 1 KM-BLOCK BOOTSTRAP INTERVAL).

Model	Unit (vehicle) SD	Segment SD	Pass residual SD	Units
REAL GBM	7.3 [6.6, 8.5]	21.7	15.2	9
SIM P79	7.7 [6.8, 9.6]	21.8	16.2	9

TABLE S8

UNIT-DISJOINT REAL BASELINE: LIGHTGBM TRAINED ONLY ON PASSES OF TELEMATICS UNITS THAT NEVER DRIVE THE HELD-OUT ROUTE, VS A SIZE-MATCHED RANDOM SUBSET OF ALL UNITS (5 DRAWS; MEAN [MIN, MAX]) AND THE FULL V1 MODEL. THE M3 FOLD KEEPS ONLY URBAN TRAINING DATA AND IS NOT INTERPRETABLE AS A VEHICLE EFFECT.

Held-out route	Train units	Classes	Unit-disjoint	Size-matched	All units (v1)
CPH1	3,176	motorway, urban	0.63	0.63 [0.60, 0.65]	0.60
CPH6	11,976	motorway, urban	0.69	0.70 [0.69, 0.71]	0.70
M3	2,924	urban	0.62	0.16 [0.14, 0.19]	0.13
M13	7,323	motorway, urban	0.15	0.15 [0.15, 0.15]	0.15

TABLE S9

APPROXIMATE ANCHOR TO BAJIC ET AL. (2021): WITHIN-ROUTE SPATIAL HOLDOUT ON M3 (EACH DIRECTION: FIRST 80 % OF THE CHAINAGE FOR TRAINING, LAST 20 % FOR TESTING). BAJIC ET AL. REPORTED MAE 0.16–0.17, RMSE 0.21 AND R^2 0.64–0.66 FOR THEIR BEST MODELS (AND MAE 0.33 FOR THE MEAN BASELINE) WITH A DIFFERENT REFERENCE SURVEY AND PROCESSING (A PROFILER OPERATED BY THE DANISH ROAD DIRECTORATE, IRI PER 10 M AVERAGED OVER OVERLAPPING 100 M WINDOWS) AND THEIR OWN PASS SELECTION, SO THE NUMBERS ARE NOT DIRECTLY COMPARABLE.

Model	MAE	RMSE	R^2
train mean (baseline)	0.319	0.391	−0.02
LightGBM	0.115	0.171	0.81
vibration index (alpha fitted)	0.133	0.185	0.77
MLP (5-seed mean prediction)	0.119	0.175	0.80

TABLE S10

SEGMENT MAE VS NUMBER OF PASSES AVERAGED PER SEGMENT, ON THE FIXED SET OF SEGMENTS WITH $\geq$8 PASSES (20 RANDOM PASS SUBSETS).

Model	Road	1	2	3	5	8
real LightGBM	motorway	0.20	0.17	0.16	0.15	0.14
real LightGBM	urban	0.66	0.60	0.59	0.57	0.56
real index (M2c)	motorway	0.33	0.33	0.32	0.32	0.32
real index (M2c)	urban	1.19	1.10	1.06	1.03	1.00
sim LightGBM (V0)	motorway	0.24	0.22	0.21	0.20	0.20
sim LightGBM (V0)	urban	0.64	0.56	0.54	0.53	0.52

TABLE S11

CALIBRATED QUARTER-CAR SPECTRAL INVERSION (M7) PER HELD-OUT ROUTE: FITTED NORMALISED QUARTER CAR (FIT TO THE EMPIRICAL TRANSMISSIBILITY OF THE THREE TRAINING ROUTES) AND SEGMENT MAE WITHOUT AND WITH THE LOG-LINEAR OUTPUT CALIBRATION.

Route	K_1	K_2	C	U	gain	fit RMSE	MAE	MAE, calibrated
CPH1	1919	192	7.3	0.29	0.63	0.41	0.68	0.70
CPH6	1903	188	7.3	0.29	0.60	0.40	0.71	0.84
M3	1791	203	7.0	0.27	0.56	0.41	0.27	0.39
M13	1876	189	7.2	0.29	0.60	0.41	0.17	0.22

TABLE S12

ARAN9000 VS P79 (100 M, MEAN OF BOTH WHEEL PATHS) AFTER HEADING-CONSTRAINED MATCHING AND REMOVAL OF ZERO-IRI RECORDS. SURVEYS: P79 10-09-2020 (CPH1, M3, M13) AND 04-06-2020 (CPH6); ARAN 12-08-2020 (CPH1), 13-08-2020 (M3, M13) AND 30-08-2021 (CPH6).

Scope	Segments	MAE	Bias	Pearson	Spearman	Median ratio	>15 m/km (%)
all	1645	0.98	+0.85	0.68	0.96	1.10	2.7
CPH1	483	1.51	+1.30	0.60	0.87	1.08	3.9
CPH6	410	1.89	+1.63	0.50	0.80	1.06	6.1
M13	230	0.12	+0.09	0.96	0.93	1.10	0.0
M3	522	0.15	+0.14	0.95	0.94	1.12	0.0
urban	893	1.68	+1.45	0.54	0.84	1.07	4.9
motorway	752	0.14	+0.12	0.95	0.95	1.12	0.0

S5. Simulator variants and paired data: additional results

Fig. S4 and Table S13 give the zero-shot results of every variant and their combination with 5 and 20 km of paired drives; Tables S14 and S15 list all variants for LightGBM and for the vibration index trained in simulation, including the full V7 and V8 sets and the game-engine corpus, and Fig. S5 shows segment-level predictions. Across V0–V8 the scale b (slope of log prediction on log IRI with route fixed effects) stays between 0.62 and 0.71 (real-trained 0.66), so the variants differ in level rather than in slope; adding synthetic to measured profiles (V10) changes little.

Table S16 summarises the paired-data comparison (Fig. 3 of the main paper) with pass counts and within-route MAEs, Table S17 gives every variant at equal profile budget and with a level offset, Table S18 with an affine map and with joint training, and Table S19 intervals over the test routes at 5 and 20 km. At equal profile budget, simulated LightGBM beats real-trained LightGBM at 1–2 km for every variant but not the real-trained vibration index (simulation better in 3–10 of 20 draws at 1 km), and a level offset from the same k km closes the gap of V4 (+0.01 m/km at 20 km) but not that of V1, V3 and V5. An affine map fitted to the paired passes was less accurate than the better of the level offset and joint training for 9 of 9 variants at 5 km and 9 of 9 at 20 km. At 20 km, joint training is worse than real-only training in 4 variant–route combinations on M3 and 1 on CPH6, and at 1 km its error still depends on the variant (0.44–0.56 m/km). The index without simulation uses the car's own speed exponent, estimated without labels within segments of the training routes ($\alpha = 0.59$–0.72; simulated vibration: 1.19–1.26), with an affine fit to the paired passes (Table S18, last row). Table S20 gives the effect of keeping fewer passes per segment, and Table S21 further combinations including the 1D-CNN.

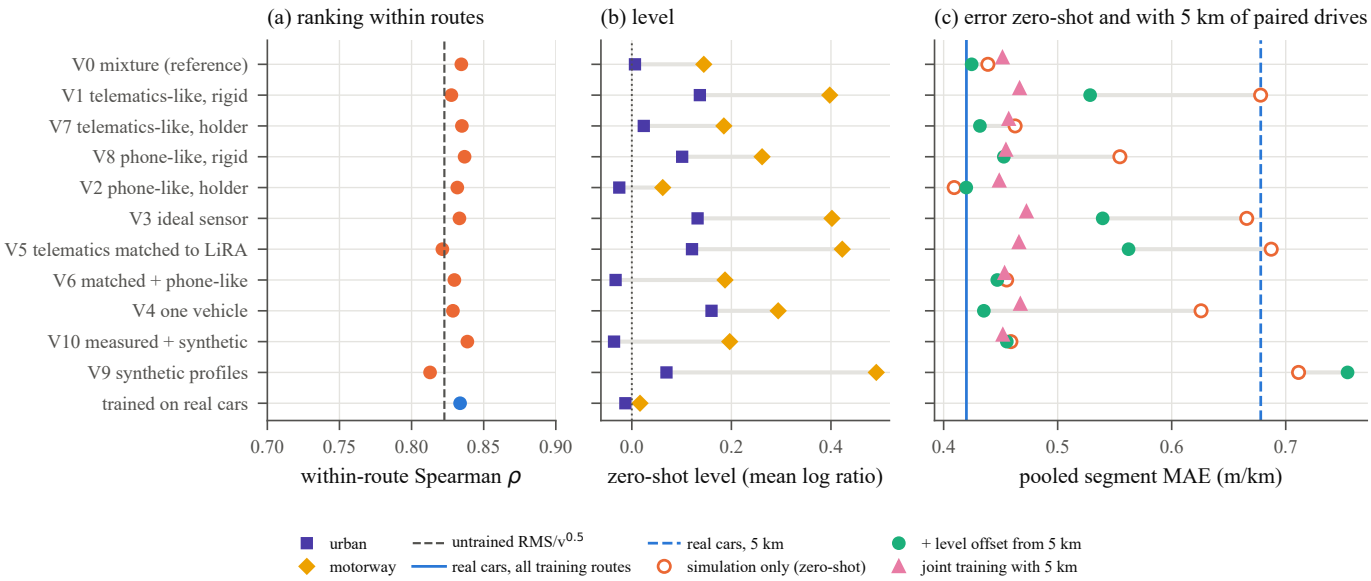


Fig. S4. Simulator variants, LightGBM trained on simulated drives only and tested on the held-out real routes. (a) Within-route Spearman ρ. (b) Zero-shot level (mean log ratio of prediction to label; mean of a class's two routes). (c) Pooled segment MAE zero-shot and with 5 km of paired drives (means over 20 draws).

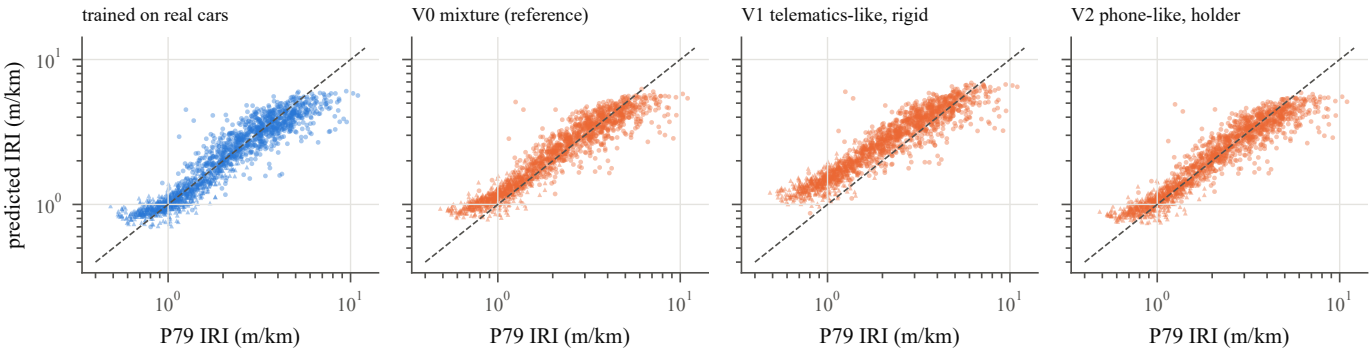


Fig. S5. Predicted vs P79 IRI on the held-out routes (segment means; circles urban, triangles motorway; log axes): LightGBM trained on real cars and trained only on simulator variants V0, V1 and V2.

TABLE S13

SIMULATOR VARIANTS: LIGHTGBM TRAINED ONLY ON SIMULATED DRIVES (ZERO-SHOT), TESTED ON THE HELD-OUT REAL ROUTES. POOLED SEGMENT MAE AND BIAS IN M/KM; LEVEL: MEAN LOG(PREDICTION/LABEL) OVER THE SEGMENTS OF A ROUTE, AVERAGED OVER THE TWO ROUTES OF A ROAD CLASS; WITHIN-ROUTE SPEARMAN ρ. RIGHT: POOLED MAE WITH 5 OR 20 KM OF PAIRED DRIVES USED FOR A LEVEL OFFSET (2.5–97.5 % RANGE OVER THE 20 DRAWS IN BRACKETS) OR FOR JOINT TRAINING; MEANS OVER 20 DRAWS. REAL-ONLY LIGHTGBM ON THE SAME DRIVES: 0.68 M/KM (5 KM), 0.46 M/KM (20 KM). SCALE b AND SPEED ELASTICITIES: TABLE S14.

	Zero-shot					+ level offset		+ joint training	
Simulator variant	MAE	bias	level urban	level motorway	ρ	5 km	20 km	5 km	20 km
V0 mixture (reference)	0.44	+0.01	+0.01	+0.14	0.83	0.42 [0.40, 0.45]	0.42	0.45	0.44
V1 telematics-like, rigid	0.68	+0.43	+0.14	+0.40	0.83	0.53 [0.47, 0.60]	0.54	0.47	0.44
V7 telematics-like + holder	0.46	+0.07	+0.02	+0.18	0.83	0.43 [0.40, 0.46]	0.43	0.46	0.44
V8 phone-like, rigid	0.55	+0.27	+0.10	+0.26	0.84	0.45 [0.42, 0.49]	0.46	0.45	0.44
V2 phone-like + holder	0.41	−0.09	−0.03	+0.06	0.83	0.42 [0.40, 0.45]	0.41	0.45	0.43
V3 ideal sensor	0.67	+0.40	+0.13	+0.40	0.83	0.54 [0.48, 0.62]	0.55	0.47	0.45
V5 telematics matched to LiRA-CD	0.69	+0.40	+0.12	+0.42	0.82	0.56 [0.51, 0.64]	0.58	0.47	0.44
V6 matched telematics + phone-like	0.46	−0.04	−0.03	+0.19	0.83	0.45 [0.41, 0.48]	0.45	0.45	0.43
V4 one mid-range vehicle	0.63	+0.40	+0.16	+0.29	0.83	0.44 [0.41, 0.46]	0.43	0.47	0.44
V10 measured + synthetic profiles	0.46	−0.04	−0.04	+0.20	0.84	0.46 [0.41, 0.50]	0.46	0.45	0.43
V9 synthetic profiles only	0.71	+0.36	+0.07	+0.49	0.81	0.75 [0.65, 0.88]	0.77	–	–
trained on real cars (M3)	0.42	−0.11	−0.01	+0.02	0.83	–	–	–	–

TABLE S14

ALL SIMULATOR VARIANTS (LIGHTGBM), INCLUDING THE FULL V7/V8 SETS (ALL PASSES INSTEAD OF A RANDOM HALF) AND THE GAME-ENGINE CORPUS WITH SMOOTHED AND UNSMOOTHED LABELS (SECTION S2). COLUMNS AS IN TABLE S13: ZERO-SHOT MAE AND BIAS, WITHIN-ROUTE SPEARMAN, SCALE b, POOLED WITHIN-SEGMENT SPEED ELASTICITY, AND MAE AFTER A LEVEL OFFSET FROM k KM OF REAL DATA OF THE TRAINING ROUTES.

Simulator variant	Zero-shot	Bias (m/km)		Within-	Scale	Speed	+ level from k km real		
	MAE	urban	motorway	route ρ	b	elasticity	1 km	5 km	20 km
V0 mixture (reference)	0.44	−0.12	+0.17	0.83	0.69	−0.06	0.49	0.42 [0.40, 0.45]	0.42
V1 telematics-like, rigid	0.68	+0.32	+0.56	0.83	0.66	+0.02	0.59	0.53 [0.47, 0.60]	0.54
V7 telematics-like + holder	0.46	−0.05	+0.23	0.83	0.70	−0.15	0.50	0.43 [0.40, 0.46]	0.43
V8 phone-like, rigid	0.55	+0.21	+0.34	0.84	0.71	−0.00	0.51	0.45 [0.42, 0.49]	0.46
V2 phone-like + holder	0.41	−0.22	+0.06	0.83	0.69	−0.10	0.48	0.42 [0.40, 0.45]	0.41
V3 ideal sensor	0.67	+0.27	+0.56	0.83	0.62	+0.01	0.60	0.54 [0.48, 0.62]	0.55
V5 telematics matched to LiRA-CD	0.69	+0.25	+0.60	0.82	0.64	+0.01	0.62	0.56 [0.51, 0.64]	0.58
V6 matched telematics + phone-like	0.46	−0.24	+0.22	0.83	0.69	+0.06	0.52	0.45 [0.41, 0.48]	0.45
V4 one mid-range vehicle	0.63	+0.39	+0.41	0.83	0.66	−0.05	0.50	0.44 [0.41, 0.46]	0.43
V10 measured + synthetic profiles	0.46	−0.26	+0.23	0.84	0.66	−0.03	0.53	0.46 [0.41, 0.50]	0.46
V9 synthetic profiles only	0.71	+0.09	+0.70	0.81	0.66	+0.23	0.84	0.75 [0.65, 0.88]	0.77
V7, all passes	0.50	+0.00	+0.30	0.84	0.68	−0.11	0.52	0.45 [0.41, 0.49]	0.46
V8, all passes	0.55	+0.19	+0.32	0.83	0.70	+0.02	0.51	0.45 [0.42, 0.49]	0.46
game engine, smoothed labels	1.36	−2.21	−0.08	0.35	0.48	+0.79	1.77	1.59 [1.30, 1.87]	1.66
game engine, unsmoothed labels	1.97	−0.47	+2.94	0.32	0.42	+1.39	2.56	2.22 [1.66, 2.85]	2.42
trained on real cars (M3)	0.42	−0.19	−0.02	0.83	0.66	−0.15	–	–	–

TABLE S15

ALL SIMULATOR VARIANTS (VIBRATION INDEX WITH THE SPEED EXPONENT FITTED IN SIMULATION), INCLUDING THE FULL V7/V8 SETS (ALL PASSES INSTEAD OF A RANDOM HALF) AND THE GAME-ENGINE CORPUS WITH SMOOTHED AND UNSMOOTHED LABELS (SECTION S2). COLUMNS AS IN TABLE S13: ZERO-SHOT MAE AND BIAS, WITHIN-ROUTE SPEARMAN, SCALE b, POOLED WITHIN-SEGMENT SPEED ELASTICITY, AND MAE AFTER A LEVEL OFFSET FROM k KM OF REAL DATA OF THE TRAINING ROUTES.

Simulator variant	Zero-shot	Bias (m/km)		Within-	Scale	Speed	+ level from k km real		
	MAE	urban	motorway	route ρ	b	elasticity	1 km	5 km	20 km
V0 mixture (reference)	0.57	+0.24	+0.28	0.81	0.75	−0.48	0.57	0.52 [0.48, 0.55]	0.52
V1 telematics-like, rigid	0.66	+0.53	+0.36	0.81	0.76	−0.49	0.56	0.51 [0.47, 0.54]	0.51
V7 telematics-like + holder	0.56	+0.24	+0.28	0.82	0.75	−0.45	0.56	0.51 [0.47, 0.55]	0.51
V8 phone-like, rigid	0.63	+0.44	+0.34	0.81	0.76	−0.47	0.58	0.52 [0.48, 0.56]	0.52
V2 phone-like + holder	0.55	+0.13	+0.21	0.81	0.77	−0.49	0.58	0.52 [0.49, 0.56]	0.53
V3 ideal sensor	0.63	+0.38	+0.40	0.81	0.72	−0.45	0.58	0.52 [0.48, 0.57]	0.53
V5 telematics matched to LiRA-CD	0.63	+0.33	+0.42	0.82	0.73	−0.42	0.58	0.52 [0.47, 0.57]	0.53
V6 matched telematics + phone-like	0.54	+0.05	+0.33	0.82	0.72	−0.40	0.59	0.53 [0.48, 0.58]	0.53
V4 one mid-range vehicle	0.74	+0.68	+0.42	0.82	0.78	−0.45	0.57	0.51 [0.47, 0.55]	0.52
V10 measured + synthetic profiles	0.57	+0.18	+0.36	0.82	0.74	−0.39	0.59	0.53 [0.47, 0.58]	0.53
V9 synthetic profiles only	0.69	−0.20	+0.74	0.83	0.65	+0.05	0.98	0.92 [0.78, 1.07]	0.95
V7, all passes	0.55	+0.12	+0.31	0.82	0.73	−0.41	0.58	0.53 [0.48, 0.58]	0.53
V8, all passes	0.63	+0.44	+0.33	0.81	0.77	−0.47	0.58	0.52 [0.48, 0.55]	0.52
game engine, smoothed labels	0.85	−1.37	+0.08	0.83	0.50	−0.11	1.04	0.97 [0.85, 1.14]	1.00
game engine, unsmoothed labels	1.96	+1.51	+2.36	0.82	0.46	+0.08	1.25	1.18 [1.02, 1.40]	1.23
trained on real cars (M3)	0.67	−0.44	+0.26	0.76	0.50	−0.64	–	–	–

TABLE S16

SIMULATION AND PAIRED REAL DRIVES ON THE SAME 20 DRAWS OF k KM OF TRAINING-ROUTE ROAD (PASSES: MEAN NUMBER OF SEGMENT-PASSES). POOLED MAE (BRACKETS: 2.5–97.5 % OF DRAWS) AND WITHIN-ROUTE MAE, IN M/KM. REAL-ONLY: LIGHTGBM OR THE THREE-PARAMETER VIBRATION INDEX ON THE PAIRED PASSES. EQUAL PROFILE BUDGET: LIGHTGBM ON V0 SIMULATED PASSES OVER THE PROFILES OF THE SAME k KM (DIFFERENCE TO REAL-ONLY LIGHTGBM; DRAWS IN WHICH SIMULATION WAS BETTER). ALL PROFILES: SIMULATION OVER ALL TRAINING-ROUTE PROFILES COMBINED WITH THE PAIRED PASSES; RANGES OVER V0–V8 OF THE MEANS, AND OF THE NUMBER OF DRAWS IN WHICH JOINT TRAINING BEAT REAL-ONLY LIGHTGBM.

		Real-only			Equal profile budget (V0)		All profiles + level offset		All profiles + joint training		
k	passes	LightGBM pooled	within	index	pooled	− real (better)	pooled	within	pooled	within	better
1	92	1.63 [1.08, 2.36]	1.572	1.30	1.33	−0.30 (20)	0.48–0.62	0.450–0.576	0.44–0.56	0.415–0.543	20
2	184	1.14 [0.63, 1.62]	1.105	0.94	0.86	−0.28 (20)	0.45–0.58	0.421–0.539	0.45–0.54	0.426–0.519	20
5	475	0.68 [0.50, 1.07]	0.629	0.72	0.60	−0.08 (16)	0.42–0.56	0.395–0.519	0.45–0.47	0.422–0.445	20
10	947	0.52 [0.45, 0.67]	0.487	0.71	0.49	−0.03 (15)	0.42–0.58	0.391–0.537	0.45–0.48	0.418–0.445	18–20
20	1,905	0.46 [0.43, 0.50]	0.433	0.69	0.47	+0.01 (9)	0.41–0.58	0.388–0.532	0.43–0.45	0.405–0.419	16–20
40	3,777	0.44 [0.42, 0.47]	0.412	0.68	0.45	+0.01 (3)	–	–	–	–	–
all		0.42	0.395	–	0.44 (zero-shot)	–	–	–	–	–	–

TABLE S17

EVERY MEASURED-PROFILE SIMULATOR VARIANT AGAINST LIGHTGBM TRAINED ON THE REAL PASSES OVER THE SAME k KM (20 PAIRED DRAWS OF TRAINING-ROUTE BLOCKS). ENTRIES: MEAN DIFFERENCE IN POOLED SEGMENT MAE, SIMULATION ARM MINUS REAL-ONLY (M/KM; NEGATIVE = SIMULATION BETTER), AND IN PARENTHESES THE NUMBER OF THE 20 DRAWS IN WHICH THE SIMULATION ARM WAS BETTER. "SAME k KM": TRAINED ON SIMULATED PASSES OVER THE PROFILES OF THE SAME BLOCKS ONLY (EQUAL PROFILE BUDGET). "ALL PROFILES + LEVEL": TRAINED ON ALL TRAINING-ROUTE PROFILES, LEVEL OFFSET FROM THE k KM OF REAL PASSES. "+ AFFINE": AFFINE CALIBRATION INSTEAD OF THE OFFSET.

	Same k km of profiles			All profiles + level from k km				+ affine	
Variant	2 km	5 km	20 km	1 km	5 km	10 km	20 km	5 km	20 km
V0	−0.28 (20)	−0.08 (16)	+0.01 (9)	−1.14 (20)	−0.25 (20)	−0.10 (20)	−0.04 (19)	−0.20 (19)	−0.03 (17)
V1	−0.09 (16)	+0.11 (2)	+0.24 (0)	−1.04 (20)	−0.15 (14)	+0.02 (5)	+0.08 (0)	−0.18 (18)	+0.02 (7)
V2	−0.27 (20)	−0.08 (16)	−0.02 (17)	−1.15 (20)	−0.26 (20)	−0.11 (20)	−0.05 (20)	−0.21 (19)	−0.03 (19)
V3	−0.16 (18)	+0.06 (4)	+0.22 (0)	−1.03 (20)	−0.14 (14)	+0.04 (4)	+0.09 (0)	−0.20 (19)	−0.01 (12)
V4	−0.17 (18)	+0.03 (6)	+0.19 (0)	−1.13 (20)	−0.24 (20)	−0.09 (20)	−0.03 (18)	−0.21 (20)	−0.02 (17)
V5	−0.12 (16)	+0.09 (3)	+0.23 (0)	−1.01 (20)	−0.12 (13)	+0.06 (3)	+0.12 (0)	−0.18 (18)	+0.02 (5)
V6	−0.24 (20)	−0.05 (13)	+0.02 (4)	−1.11 (20)	−0.23 (20)	−0.07 (17)	−0.01 (14)	−0.21 (19)	−0.02 (16)
V7	−0.22 (19)	−0.03 (10)	+0.05 (0)	−1.13 (20)	−0.25 (20)	−0.09 (20)	−0.03 (18)	−0.20 (18)	−0.01 (13)
V8	−0.21 (18)	−0.01 (9)	+0.10 (0)	−1.12 (20)	−0.23 (20)	−0.06 (18)	−0.00 (14)	−0.20 (19)	−0.02 (14)
Real-only MAE	1.14	0.68	0.46	1.63	0.68	0.52	0.46	0.68	0.46

TABLE S18

EVERY SIMULATOR VARIANT COMBINED WITH k KM OF PAIRED DRIVES THROUGH AN AFFINE MAP (FITTED ON THE PAIRED PASSES) OR BY JOINT TRAINING (SIMULATED UNITS AND PAIRED PASSES, EQUAL TOTAL WEIGHT), AGAINST REAL-ONLY LIGHTGBM ON THE SAME PASSES (20 PAIRED DRAWS). ENTRIES: MEAN DIFFERENCE IN POOLED SEGMENT MAE (M/KM; NEGATIVE = SIMULATION BETTER) AND, IN PARENTHESES, THE NUMBER OF DRAWS IN WHICH SIMULATION WAS BETTER. LAST ROW: POOLED MAE OF A NO-SIMULATOR INDEX $\log \mathrm{IRI} = c + b \log(\mathrm{RMS}/v^{\alpha})$ WITH α THE CAR'S LABEL-FREE WITHIN-SEGMENT SPEED EXPONENT (0.59–0.72), AFFINE-CALIBRATED ON THE SAME PASSES.

	+ affine map				+ joint training			
Variant	1 km	5 km	10 km	20 km	1 km	5 km	10 km	20 km
V0	−0.42 (17)	−0.20 (19)	−0.07 (19)	−0.03 (17)	−1.17 (20)	−0.23 (20)	−0.07 (19)	−0.03 (20)
V1	−0.45 (18)	−0.18 (18)	−0.03 (11)	+0.02 (7)	−1.08 (20)	−0.21 (20)	−0.06 (18)	−0.02 (17)
V2	−0.42 (18)	−0.21 (19)	−0.08 (19)	−0.03 (19)	−1.19 (20)	−0.23 (20)	−0.08 (20)	−0.03 (20)
V3	−0.25 (18)	−0.20 (19)	−0.05 (14)	−0.01 (12)	−1.07 (20)	−0.21 (20)	−0.05 (18)	−0.01 (16)
V4	−0.71 (19)	−0.21 (20)	−0.07 (19)	−0.02 (17)	−1.08 (20)	−0.21 (20)	−0.06 (19)	−0.02 (18)
V5	−0.60 (18)	−0.18 (18)	−0.02 (11)	+0.02 (5)	−1.07 (20)	−0.21 (20)	−0.05 (19)	−0.02 (18)
V6	−0.61 (18)	−0.21 (19)	−0.07 (17)	−0.02 (16)	−1.15 (20)	−0.22 (20)	−0.07 (19)	−0.03 (19)
V7	−0.55 (18)	−0.20 (18)	−0.05 (16)	−0.01 (13)	−1.16 (20)	−0.22 (20)	−0.06 (19)	−0.02 (18)
V8	−0.46 (18)	−0.20 (19)	−0.06 (17)	−0.02 (14)	−1.13 (20)	−0.22 (20)	−0.06 (19)	−0.02 (18)
V10	−0.41 (18)	−0.20 (19)	−0.08 (19)	−0.03 (17)	−1.15 (20)	−0.23 (20)	−0.07 (19)	−0.03 (20)
Real-only LightGBM (MAE)	1.63	0.68	0.52	0.46	1.63	0.68	0.52	0.46
No-simulator index (MAE)	1.09	0.78	0.82	0.76				

TABLE S19

TEST-ROUTE UNCERTAINTY OF THE CALIBRATION COMPARISON: MEAN OVER THE 20 DRAWS OF THE POOLED-MAE DIFFERENCE TO REAL-ONLY LIGHTGBM ON THE SAME k KM (M/KM), WITH A 95 % INTERVAL FROM A BOOTSTRAP OVER 5 KM BLOCKS OF THE TEST SEGMENTS (40 BLOCKS). THE INTERVAL CONDITIONS ON THE 20 CALIBRATION DRAWS.

	+ level offset		+ joint training	
Variant	5 km	20 km	5 km	20 km
V0	−0.25 [−0.29, −0.22]	−0.04 [−0.05, −0.03]	−0.23 [−0.26, −0.19]	−0.03 [−0.03, −0.02]
V1	−0.15 [−0.18, −0.12]	+0.08 [+0.04, +0.11]	−0.21 [−0.26, −0.17]	−0.02 [−0.03, −0.01]
V2	−0.26 [−0.30, −0.22]	−0.05 [−0.06, −0.03]	−0.23 [−0.26, −0.20]	−0.03 [−0.04, −0.02]
V3	−0.14 [−0.17, −0.11]	+0.09 [+0.05, +0.13]	−0.21 [−0.25, −0.16]	−0.01 [−0.03, −0.00]
V4	−0.24 [−0.28, −0.21]	−0.03 [−0.05, −0.01]	−0.21 [−0.26, −0.16]	−0.02 [−0.03, −0.01]
V5	−0.12 [−0.15, −0.09]	+0.12 [+0.08, +0.16]	−0.21 [−0.26, −0.17]	−0.02 [−0.03, −0.01]
V6	−0.23 [−0.26, −0.20]	−0.01 [−0.04, +0.01]	−0.22 [−0.26, −0.19]	−0.03 [−0.04, −0.02]
V7	−0.25 [−0.29, −0.21]	−0.03 [−0.04, −0.01]	−0.22 [−0.26, −0.18]	−0.02 [−0.03, −0.01]
V8	−0.23 [−0.25, −0.20]	−0.00 [−0.02, +0.01]	−0.22 [−0.27, −0.18]	−0.02 [−0.03, −0.01]

TABLE S20

PASS DENSITY OF THE PAIRED DATA AT $k = 5$ KM: EACH SEGMENT KEEPS AT MOST m OF ITS REAL PASSES (RANDOM, INDEPENDENT OF THE LABELS; 20 DRAWS). POOLED MAE IN M/KM; FOR THE SIMULATION ARMS THE RANGE OVER V0–V8 AND, IN PARENTHESES, THE RANGE OVER THE VARIANTS OF THE NUMBER OF DRAWS IN WHICH THE ARM BEAT REAL-ONLY LIGHTGBM WITH THE SAME PASSES.

m	segment-passes	real-only LightGBM	+ level offset	+ joint training
1	49	0.93	0.43–0.55 (20)	0.44–0.49 (20)
2	98	0.75	0.42–0.55 (18–20)	0.44–0.48 (20)
5	242	0.69	0.42–0.55 (13–20)	0.45–0.48 (20)
all	475	0.68	0.42–0.56 (13–20)	0.45–0.47 (20)

TABLE S21

FURTHER COMBINATIONS OF SIMULATION AND PAIRED DRIVES, INCLUDING THE 1D-CNN (5 DRAWS PER k, SIMULATION SET V10 WITH MEASURED AND SYNTHETIC PROFILES; POOLED SEGMENT MAE IN M/KM, MEAN OVER DRAWS). REAL-ONLY: TRAINED ON THE PAIRED DRIVES ALONE; SIM INDEX + REAL INTERCEPT: VIBRATION INDEX FITTED IN SIMULATION WITH A LEVEL OFFSET; SIM+REAL LIGHTGBM: JOINT TRAINING; SIM-PRETRAINED 1D-CNN: FINE-TUNED ON THE PAIRED DRIVES (THREE SEEDS; IT DIVERGES ON SOME DRAWS AT SMALL k).

Arm	1	2	5	10	20	all
real-only 1D-CNN	>50	–	1.25	–	0.57	0.55
real-only LightGBM	1.41	1.18	0.82	0.50	0.46	0.42
real-only physics index	1.10	0.93	0.81	0.71	0.67	0.67
sim physics index + real intercept	0.53	0.52	0.54	0.54	0.54	0.53
sim+real LightGBM (balanced weights)	0.46	0.46	0.45	0.44	0.43	0.41
sim-pretrained 1D-CNN + real fine-tuning	>50	–	0.78	–	0.53	0.56

S6. Spectra and speed: additional results

Table S22 gives the band means of the spectral differences in Fig. 2a, Table S23 the 10–15 Hz ratio by road class and the speed scaling of raw vibration in simulation and reality, Table S24 the association of every band's amplitude ratio with the zero-shot bias and level (R^2 0.71 for 10–15 Hz and 0.78 for the overall RMS), and Table S25 the band-limited retraining. The 10–15 Hz deficit is larger on motorway than on urban segments for 9 of 9 variants, as is the level error. Passes logged at 333 Hz contain more, not less, 10–15 Hz power than passes logged at 52.6 Hz on the same cells (52.6 minus 333 Hz: -0.24 in $\log_{10}$, confounded with the device), so the excess is not an aliasing artefact; in the 333 Hz passes, 25 % of the vibration energy above 0.3 Hz lies at 10–20 Hz, 21 % above 20 Hz and 13 % above the 26.3 Hz Nyquist frequency of the slower logging (medians over 5,721 units).

Speed on a fixed road. Each segment is driven about ten times at different speeds (median within-segment SD of log speed 0.12), so the speed dependence of an estimate can be measured on a road of constant roughness: the within-segment elasticity is the slope of the log estimate on log speed with segment fixed effects, and ± 10 % between 40 and 100 km/h corresponds to |elasticity| $\leq$0.104 (Fig. S6, Table S26; 1 km-block intervals for single elasticities). The real car's vibration RMS rises by $+83$ % between 40 and 100 km/h (elasticity $+0.66$ [$+0.60$, $+0.71$]). The estimators trained on real cars over-correct for it: LightGBM's estimate changes by -13 % over the same range (-0.15 [-0.19, -0.11]), also without speed inputs (-0.14), the MLP and CNN behave alike (-0.19 and -0.19), and the vibration index over-corrects most (-0.64). V0-trained LightGBM changes by -6 % (-0.06 [-0.11, -0.01]); paired on the same passes, its elasticity exceeds that of the real-trained model by $+0.09$ [$+0.06$, $+0.12$] (5 km blocks), entirely on urban roads ($+0.12$ [$+0.09$, $+0.15$]; motorways $+0.02$ [-0.06, $+0.06$]), and across the nine variants the paired difference ranges from $+0.00$ to $+0.21$ (interval excluding zero for 8 of 9). These elasticities change little with telematics-unit fixed effects or with the within-pass speed SD as a covariate (Table S27), except that slow urban passes are more dynamic (within-segment correlation of log speed and log speed SD -0.50): with the speed SD as a covariate the real-trained urban elasticity shrinks to -0.06 [-0.16, $+0.02$], while the paired urban difference stays at $+0.12$ [$+0.09$, $+0.16$]. The speed coefficient of variation, tied to $1/v$ by construction (correlation -0.86 with log speed), absorbs most of the speed variation within urban segments and turns that difference negative (-0.07 [-0.21, $+0.01$]); on urban roads speed and speed variability cannot be fully separated, so the smaller urban over-correction of simulation-trained LightGBM is not established.

TABLE S22
MEAN $\log_{10}$ PSD DIFFERENCE BY BAND, REAL MINUS SIMULATED (RE-SIMULATED PASSES OF THE FIRST 10 TRAINING VEHICLES OVER ALL P79 PROFILES), ON COMMON SEGMENT $\times$ 10 KM/H CELLS; LAST ROW: REAL PASSES LOGGED AT 52.6 HZ MINUS 333 HZ (ALL UNITS, DEVICE-CONFOUNDED).

Comparison	0.5–2 Hz	2–5 Hz	5–10 Hz	10–15 Hz	15–20 Hz	Cells
all units: 52.6Hz-333Hz	−0.06	−0.09	−0.16	−0.24	−0.40	1,765
real-V1	−0.18	+0.38	+0.01	+0.69	+0.89	3,577
real-V2	−0.20	+0.39	−0.14	+0.27	+0.06	3,716
real-V3	−0.19	+0.49	+0.08	+0.71	+0.97	3,455
real-V5	−0.20	+0.55	+0.12	+0.69	+0.91	3,127
real-V7	−0.18	+0.43	−0.01	+0.47	+0.48	3,574
real-V8	−0.15	+0.37	−0.05	+0.45	+0.48	3,277

TABLE S23
SIMULATED VS REAL VIBRATION ON THE SAME SEGMENTS, PER VARIANT. LEFT: REAL/SIMULATED AMPLITUDE RATIO OF THE 10–15 HZ BAND (CELL MEANS OVER SEGMENT $\times$ 10 KM/H SPEED BAND; ALL ROUTES, URBAN, MOTORWAY) NEXT TO THE ZERO-SHOT BIAS. RIGHT: SPEED SCALING OF THE RAW VIBRATION RMS ON A FIXED ROAD (WITHIN-SEGMENT ELASTICITY; SIMULATED PASSES RESTRICTED TO THE SPEED RANGE OF THE REAL PASSES OVER THE SAME SEGMENT; 95 % CI FROM A 5 KM-BLOCK BOOTSTRAP).

	10–15 Hz ratio real/sim			Zero-shot	RMS speed elasticity	
Source	all	urban	motorway	bias	urban	motorway
V0	1.86	1.57	2.14	+0.01	1.25 [1.17, 1.34]	1.10 [1.02, 1.18]
V1	2.41	1.98	2.81	+0.43	1.22 [1.13, 1.31]	1.07 [1.01, 1.17]
V2	1.36	1.33	1.39	−0.09	1.30 [1.19, 1.42]	1.29 [1.20, 1.34]
V3	2.37	1.92	2.81	+0.40	1.16 [1.10, 1.23]	1.40 [1.31, 1.46]
V4	1.93	1.67	2.19	+0.40	1.11 [1.07, 1.15]	1.26 [1.12, 1.41]
V5	2.26	1.93	2.60	+0.40	0.92 [0.84, 1.01]	1.32 [1.21, 1.41]
V6	1.80	1.49	2.12	−0.04	0.95 [0.85, 1.06]	1.17 [1.07, 1.25]
V7	1.80	1.57	2.00	+0.07	1.27 [1.14, 1.39]	1.27 [1.12, 1.35]
V8	1.89	1.60	2.15	+0.27	1.05 [0.97, 1.10]	1.21 [1.08, 1.31]
Real car	1	1	1	–	0.60 [0.50, 0.69]	0.85 [0.74, 0.93]

TABLE S24

ASSOCIATION, OVER THE NINE MEASURED-PROFILE VARIANTS V0–V8, BETWEEN THE LOG REAL/SIMULATED AMPLITUDE RATIO OF EACH VIBRATION FEATURE (SAME SEGMENTS AND SPEED BANDS) AND THE ZERO-SHOT POOLED BIAS OR THE MEAN LOG LEVEL: SPEARMAN AND PEARSON CORRELATIONS AND R^2 OF A LINEAR FIT. THE VARIANTS ARE NOT INDEPENDENT (V0 MIXES V1 AND V2; V6 SHARES V2'S PHONE-LIKE SENSOR).

	vs bias (m/km)			vs mean log level		
Feature	Spearman	Pearson	R^2	Spearman	Pearson	R^2
0.5–2 Hz	0.22	0.32	0.10	0.08	0.22	0.05
2–5 Hz	0.25	0.44	0.19	0.33	0.41	0.17
5–10 Hz	0.80	0.81	0.66	0.80	0.86	0.73
10–15 Hz	0.97	0.84	0.71	0.90	0.90	0.81
overall RMS	0.88	0.88	0.78	0.83	0.88	0.77

TABLE S25

BAND-LIMITED ZERO-SHOT RETRAINING (LIGHTGBM, FIRST CONFIGURATION) FOR V0–V8 WITH THREE FEATURE SETS: ALL 13 FEATURES; ALL BUT THE 10–15 HZ BAND; ONLY FEATURES WITHOUT CONTENT ABOVE 10 HZ (LOG MEAN SPEED, SPEED VARIATION, BAND RMS 0.5–2, 2–5, 5–10 HZ). LEVEL: MEAN LOG RATIO, AVERAGED OVER THE TWO ROUTES OF A ROAD CLASS; POOLED MAE ZERO-SHOT AND AFTER A LEVEL OFFSET FROM 5 KM OF PAIRED DRIVES (20 DRAWS); WITHIN-ROUTE ρ. LAST ROWS: SPREAD (MAX − MIN) OVER THE EIGHT SENSOR VARIANTS (ALL BUT V4).

	all 13					all but 10–15 Hz					below 10 Hz				
Variant	lvl urb.	lvl mot.	MAE	+offset	ρ	lvl urb.	lvl mot.	MAE	+offset	ρ	lvl urb.	lvl mot.	MAE	+offset	ρ
V0	+0.01	+0.14	0.44	0.42	0.83	+0.01	+0.16	0.44	0.43	0.83	−0.09	+0.14	0.49	0.51	0.80
V1	+0.14	+0.40	0.68	0.53	0.83	+0.14	+0.40	0.69	0.53	0.83	−0.10	+0.19	0.51	0.55	0.81
V2	−0.03	+0.06	0.41	0.42	0.83	−0.02	+0.07	0.41	0.42	0.83	−0.05	+0.13	0.48	0.49	0.80
V3	+0.13	+0.40	0.67	0.54	0.83	+0.13	+0.40	0.67	0.54	0.83	−0.04	+0.24	0.53	0.56	0.80
V4	+0.16	+0.29	0.63	0.44	0.83	+0.16	+0.31	0.64	0.44	0.83	+0.10	+0.33	0.62	0.53	0.80
V5	+0.12	+0.42	0.69	0.56	0.82	+0.13	+0.43	0.70	0.56	0.82	−0.09	+0.24	0.53	0.57	0.80
V6	−0.03	+0.19	0.46	0.45	0.83	−0.03	+0.20	0.46	0.45	0.83	−0.09	+0.20	0.53	0.55	0.79
V7	+0.02	+0.18	0.46	0.43	0.83	+0.03	+0.21	0.48	0.43	0.83	−0.12	+0.18	0.52	0.53	0.80
V8	+0.10	+0.26	0.55	0.45	0.84	+0.11	+0.27	0.56	0.45	0.83	−0.08	+0.12	0.47	0.48	0.80
spread, 8 sensor variants	0.17	0.36				0.17	0.36				0.08	0.12			

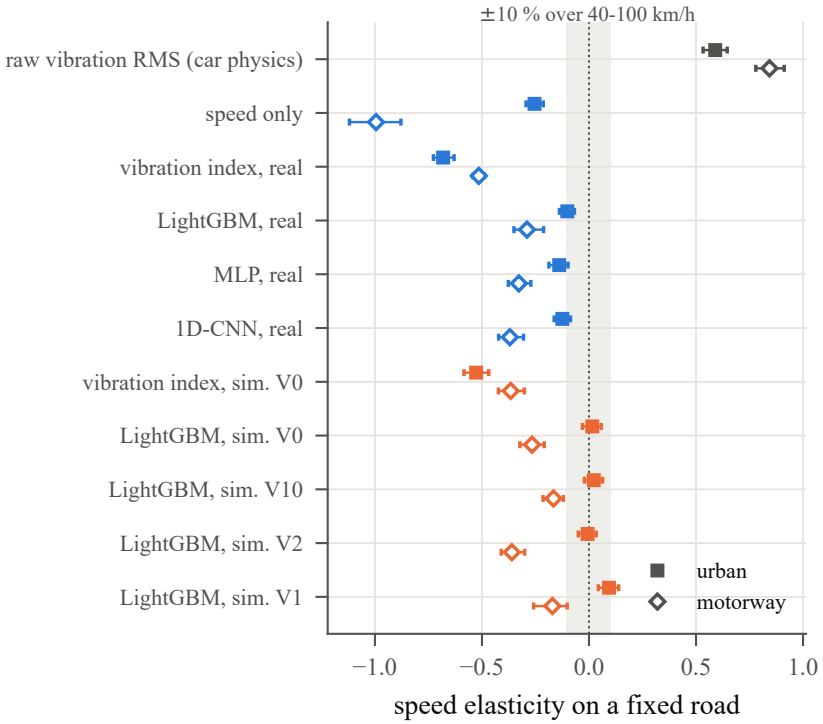


Fig. S6. Speed elasticity of each estimate on a fixed real road (slope of log estimate on log speed with segment fixed effects; 95 % intervals from a 1 km-block bootstrap; urban and motorway segments). Grey band: ±10 % change between 40 and 100 km/h. Top row: raw vibration RMS of the real car.

TABLE S26

SPEED ELASTICITY OF EACH ESTIMATE ON A FIXED ROAD: SLOPE OF LOG(PREDICTION) ON LOG(SPEED) WITH SEGMENT FIXED EFFECTS (SEGMENTS WITH $\geq$3 PASSES), 95 % CI BY 1 KM-BLOCK BOOTSTRAP; "BETWEEN": DESIGN WITH ROUTE EFFECTS AND LOG IRI AS COVARIATE. |ELASTICITY| $\leq$0.104 CORRESPONDS TO THE $\pm$10 % CRITERION OVER 40–100 KM/H. LAST COLUMN: PAIRED DIFFERENCE TO THE REAL-TRAINED ESTIMATOR OF THE SAME FAMILY (LIGHTGBM OR INDEX) ON THE SAME PASSES, 95 % CI BY 5 KM-BLOCK BOOTSTRAP (POOLED).

Estimate	Pooled	Urban	Motorway	Between	− real-trained
RAW RMS	+0.66 [+0.60, +0.71]	+0.59 [+0.53, +0.65]	+0.84 [+0.78, +0.91]	+0.64	–
REAL SPEED	−0.46 [−0.55, −0.37]	−0.25 [−0.29, −0.21]	−0.99 [−1.12, −0.88]	−0.46	–
REAL INDEX	−0.64 [−0.68, −0.59]	−0.68 [−0.73, −0.63]	−0.51 [−0.54, −0.49]	−0.65	–
REAL RMS index	+0.21 [+0.18, +0.24]	+0.20 [+0.16, +0.24]	+0.26 [+0.24, +0.28]	+0.21	–
REAL GBM	−0.15 [−0.19, −0.11]	−0.10 [−0.14, −0.07]	−0.29 [−0.35, −0.21]	−0.16	–
REAL GBM nospeed	−0.14 [−0.18, −0.10]	−0.21 [−0.24, −0.18]	+0.02 [−0.07, +0.18]	−0.17	–
REAL MLP	−0.19 [−0.23, −0.15]	−0.14 [−0.19, −0.10]	−0.33 [−0.38, −0.27]	−0.19	–
REAL CNN	−0.19 [−0.24, −0.15]	−0.12 [−0.16, −0.09]	−0.37 [−0.42, −0.31]	−0.19	–
SIM V0 GBM	−0.06 [−0.11, −0.01]	+0.02 [−0.03, +0.06]	−0.27 [−0.32, −0.21]	−0.08	+0.09 [+0.06, +0.12]
SIM V0 INDEX	−0.48 [−0.53, −0.44]	−0.53 [−0.58, −0.47]	−0.37 [−0.42, −0.30]	−0.50	+0.15 [+0.12, +0.18]
SIM V1 GBM	+0.02 [−0.03, +0.07]	+0.09 [+0.04, +0.14]	−0.17 [−0.26, −0.10]	−0.01	+0.17 [+0.13, +0.20]
SIM V1 INDEX	−0.49 [−0.54, −0.44]	−0.54 [−0.59, −0.48]	−0.36 [−0.42, −0.30]	−0.51	+0.15 [+0.11, +0.17]
SIM V2 GBM	−0.10 [−0.16, −0.05]	−0.01 [−0.05, +0.03]	−0.36 [−0.41, −0.30]	−0.11	+0.05 [+0.01, +0.08]
SIM V2 INDEX	−0.49 [−0.55, −0.44]	−0.55 [−0.61, −0.49]	−0.35 [−0.40, −0.28]	−0.52	+0.14 [+0.11, +0.17]
SIM V3 GBM	+0.01 [−0.04, +0.05]	+0.09 [+0.05, +0.13]	−0.22 [−0.29, −0.16]	−0.02	+0.16 [+0.12, +0.19]
SIM V3 INDEX	−0.45 [−0.50, −0.40]	−0.51 [−0.56, −0.45]	−0.30 [−0.36, −0.24]	−0.48	+0.19 [+0.15, +0.21]
SIM V4 GBM	−0.05 [−0.10, −0.00]	+0.03 [−0.01, +0.07]	−0.26 [−0.32, −0.21]	−0.05	+0.10 [+0.08, +0.13]
SIM V4 INDEX	−0.45 [−0.50, −0.40]	−0.50 [−0.56, −0.44]	−0.30 [−0.36, −0.24]	−0.47	+0.19 [+0.14, +0.22]
SIM V5 GBM	+0.01 [−0.03, +0.05]	+0.08 [+0.04, +0.12]	−0.16 [−0.25, −0.10]	−0.01	+0.17 [+0.13, +0.20]
SIM V5 INDEX	−0.42 [−0.47, −0.37]	−0.48 [−0.54, −0.43]	−0.24 [−0.30, −0.18]	−0.44	+0.22 [+0.19, +0.25]
SIM V6 GBM	+0.06 [+0.01, +0.11]	+0.16 [+0.11, +0.20]	−0.19 [−0.27, −0.12]	+0.05	+0.21 [+0.18, +0.24]
SIM V6 INDEX	−0.40 [−0.45, −0.35]	−0.45 [−0.50, −0.39]	−0.26 [−0.31, −0.19]	−0.42	+0.24 [+0.21, +0.26]
SIM V7 GBM	−0.15 [−0.20, −0.11]	−0.10 [−0.16, −0.06]	−0.28 [−0.33, −0.22]	−0.18	+0.00 [−0.04, +0.03]
SIM V7 INDEX	−0.45 [−0.50, −0.40]	−0.50 [−0.56, −0.44]	−0.33 [−0.39, −0.26]	−0.47	+0.18 [+0.15, +0.21]
SIM V8 GBM	−0.00 [−0.05, +0.05]	+0.08 [+0.03, +0.13]	−0.22 [−0.28, −0.15]	−0.01	+0.15 [+0.11, +0.18]
SIM V8 INDEX	−0.47 [−0.52, −0.42]	−0.52 [−0.58, −0.46]	−0.33 [−0.39, −0.26]	−0.49	+0.17 [+0.13, +0.20]
SIM V9 GBM	+0.23 [+0.19, +0.27]	+0.24 [+0.19, +0.28]	+0.21 [+0.16, +0.27]	+0.22	+0.38 [+0.33, +0.43]
SIM V9 INDEX	+0.05 [−0.01, +0.10]	−0.01 [−0.07, +0.04]	+0.21 [+0.15, +0.27]	+0.02	+0.68 [+0.65, +0.71]
SIM V10 GBM	−0.03 [−0.07, +0.01]	+0.02 [−0.02, +0.06]	−0.17 [−0.21, −0.12]	−0.05	+0.12 [+0.10, +0.15]
SIM V10 INDEX	−0.39 [−0.44, −0.34]	−0.44 [−0.50, −0.38]	−0.25 [−0.31, −0.19]	−0.41	+0.25 [+0.21, +0.27]
SIM B corr GBM	+0.79 [+0.67, +0.91]	+0.76 [+0.60, +0.91]	+0.88 [+0.75, +0.98]	+0.65	+0.94 [+0.77, +1.11]
SIM B corr INDEX	−0.11 [−0.15, −0.08]	−0.16 [−0.20, −0.12]	+0.00 [−0.04, +0.05]	−0.13	+0.52 [+0.50, +0.54]

TABLE S27

ROBUSTNESS OF THE FIXED-ROAD SPEED ELASTICITIES: (A) DESIGN OF THE MAIN PAPER; (B) WITH TELEMATICS-UNIT FIXED EFFECTS (POINT ESTIMATE); (C) WITH THE LOG WITHIN-PASS SPEED SD AS A COVARIATE; (D) WITH THE SPEED COEFFICIENT OF VARIATION AS A COVARIATE, WHICH IS MECHANICALLY TIED TO $1/v$ (WITHIN-SEGMENT CORRELATION WITH LOG SPEED -0.86 ON URBAN ROADS) AND ABSORBS MOST OF THE WITHIN-SEGMENT SPEED VARIATION. 95 % INTERVALS FROM 5 KM BLOCKS. WITHIN URBAN SEGMENTS, LOG SPEED AND LOG SPEED SD CORRELATE AT -0.50: SLOW URBAN PASSES ARE MORE DYNAMIC.

Class	Quantity	(a) main design	(b) + unit FE	(c) + log speed SD	(d) + speed CV
urban	raw vibration RMS	+0.60 [+0.51, +0.69]	+0.59	+0.65 [+0.49, +0.84]	+0.83 [+0.47, +1.40]
	real-trained LightGBM	−0.10 [−0.17, −0.05]	−0.10	−0.06 [−0.16, +0.02]	−0.08 [−0.23, +0.13]
	V0 LightGBM (zero-shot)	+0.02 [−0.05, +0.08]	+0.02	+0.06 [−0.07, +0.17]	−0.15 [−0.28, +0.03]
	V0 minus real-trained	+0.12 [+0.09, +0.15]	+0.12	+0.12 [+0.09, +0.16]	−0.07 [−0.21, +0.01]
motorway	raw vibration RMS	+0.85 [+0.73, +0.92]	+0.83	+0.86 [+0.74, +0.96]	+0.91 [+0.75, +1.03]
	real-trained LightGBM	−0.29 [−0.34, −0.14]	−0.31	−0.24 [−0.30, −0.14]	−0.18 [−0.27, −0.02]
	V0 LightGBM (zero-shot)	−0.27 [−0.32, −0.21]	−0.27	−0.22 [−0.28, −0.17]	−0.23 [−0.29, −0.13]
	V0 minus real-trained	+0.02 [−0.08, +0.06]	+0.04	+0.01 [−0.07, +0.05]	−0.05 [−0.14, +0.00]

S7. CONTROLLED SIMULATION: OTHER VEHICLES AND SENSORS

LiRA-CD contains one car model, so transfer to other vehicles was examined in simulation with freshly simulated test data (Table S28, Fig. S7). With the amount of training data fixed at 1,200 passes, the MAE on 30 unseen vehicles exceeds that on the training vehicles by +70 % with 3 training vehicles, +16 % with 30 (the number used in V0) and +6 % with 100. The upper 95 % bound of this increase is at most 10 % from 100 vehicles on with a segment bootstrap but not with a route bootstrap (upper bounds 11 and 12 % for 100 and 300 vehicles). The gap closes from both sides: the error on unseen vehicles falls (1.06 to 0.87 m/km from 3 to 100 vehicles) and the error on training vehicles rises as each gets fewer passes (0.62 to 0.82 m/km). At $N = 30$, a vehicle's error is only weakly related to its parameter distance from the nearest training vehicle (Spearman 0.14 over 30 vehicles). Sensor perturbations at test time (Table S28, right; Fig. S8) reproduce in simulation what the variants showed on the real car: the sensor path moves the level. A holder resonance at 15 Hz turns a bias of −0.65 into +0.37 m/km; 25 Hz sampling and a speed error of +20 % are the most harmful perturbations (MAE 1.39 and 1.06 m/km),

TABLE S28

CONTROLLED SIMULATION (NO REAL DATA). LEFT: NUMBER OF TRAINING VEHICLES AT A FIXED 1,200 TRAINING PASSES (HELD-OUT SYNTHETIC ROUTES, 30 UNSEEN AND UP TO 30 SEEN VEHICLES; LIGHTGBM). Δ: UNSEEN MINUS SEEN MAE RELATIVE TO SEEN, WITH THE UPPER 95 % BOUND FROM A SEGMENT AND FROM A ROUTE BOOTSTRAP; CRITERION: UPPER BOUND $\leq$10 %. RIGHT: ONE SENSOR PROPERTY CHANGED AT TEST TIME (LIGHTGBM TRAINED ON SYNTHETIC PROFILES WITH THE SENSOR MIXTURE, TESTED ON 12 HELD-OUT SYNTHETIC ROUTES DRIVEN BY 6 UNSEEN VEHICLES WITH THE TELEMATICS-LIKE SENSOR; MAE AND BIAS IN M/KM).

Vehicles N	Seen MAE	Unseen MAE	Δ (%)	Upper bound (%) segments / routes	Criterion met segments / routes
3	0.62	1.06	+70	76 / 87	no / no
10	0.75	0.92	+24	27 / 31	no / no
30	0.78	0.90	+16	18 / 21	no / no
100	0.82	0.87	+6	8 / 11	yes / no
300	0.81	0.87	+7	9 / 12	yes / no

Sensor property at test time	MAE	Bias
telematics-like, rigid (reference)	0.84	−0.65
+ holder resonance 40 Hz	0.79	−0.55
+ holder resonance 25 Hz	0.73	−0.36
+ holder resonance 15 Hz	0.88	+0.37
sampling rate 200 Hz	0.77	−0.48
sampling rate 50 Hz	0.87	−0.71
sampling rate 25 Hz	1.39	−1.36
speed error +20 %	1.06	−0.98
speed error −20 %	0.73	−0.15
white noise ×8	0.82	−0.58
mounting tilt s.d. 20°	0.84	−0.64
40 % samples lost	0.94	−0.81

white noise, tilt and lost samples the least.

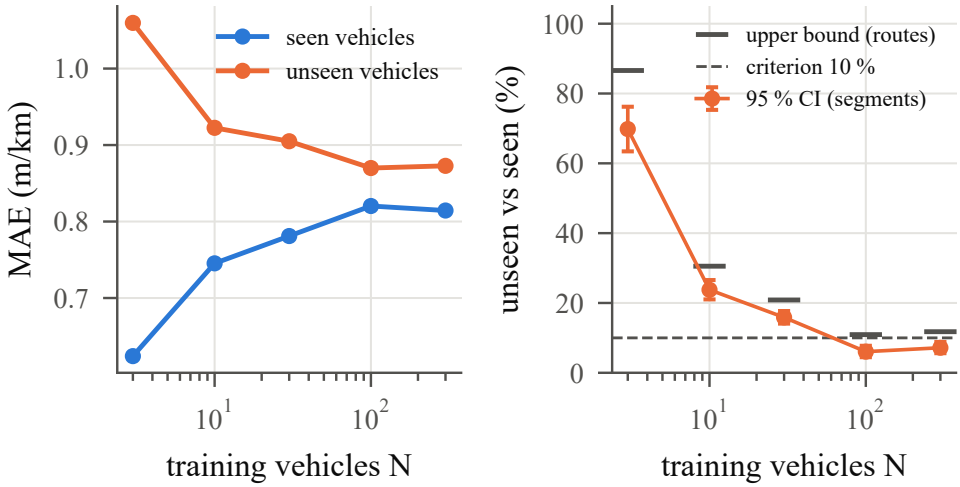


Fig. S7. Number of training vehicles at a fixed 1,200 training passes (simulation only). Left: MAE on seen and unseen vehicles. Right: relative increase from seen to unseen vehicles with 95 % interval (segment bootstrap) and the upper bound of a route bootstrap; dashed: 10 % criterion.

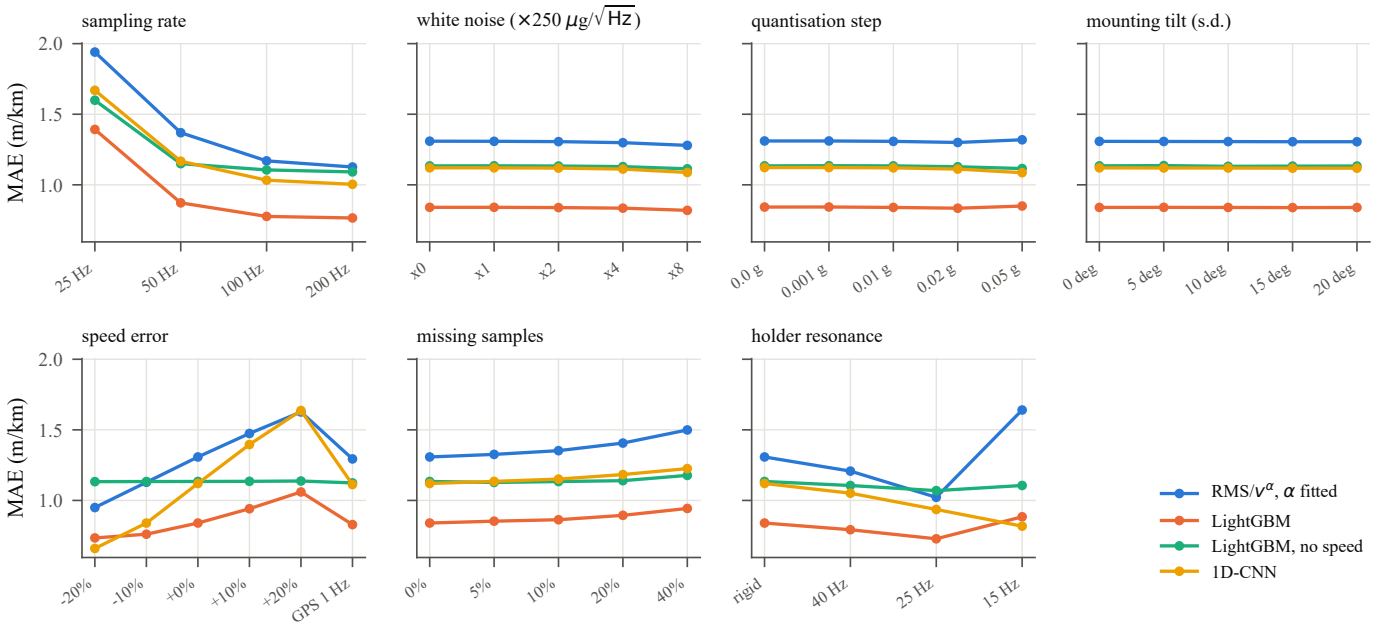


Fig. S8. Sensor robustness in simulation: estimators trained on synthetic profiles, evaluated on 12 held-out synthetic routes driven by 6 unseen vehicles with the telematics sensor configuration; one property changed at a time (common random numbers).

S8. Reproduction

Every number in the paper and this supplement is a LaTeX macro generated from the result files (`make_numbers.py`, `make_numbers_v2.py`, `make_numbers_v3.py`); tables and figures are generated by `make_tables*.py` and `make_figures*.py`. The complete pipeline is `run_all_experiments.py`; `README.md` lists the commands, the pinned environment (`environment_lock.txt`), seeds and runtimes. The code, configurations and predictions are not yet public; they will be released with the published version of the paper.